

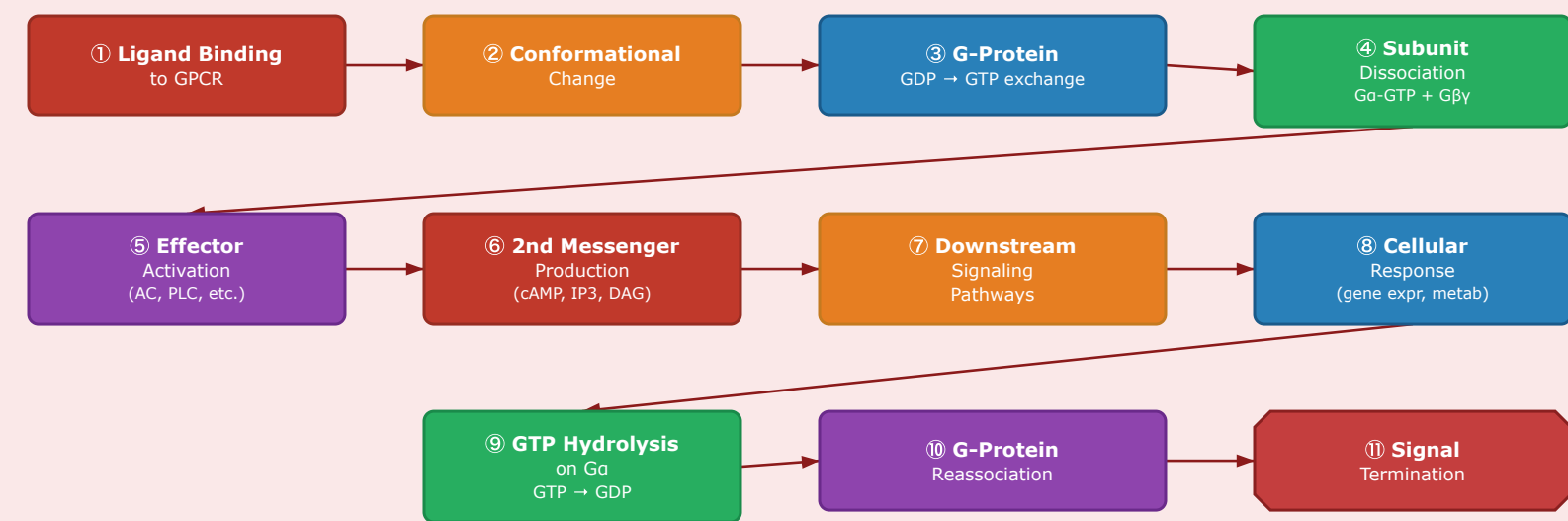
GPCR SIGNALING

GPCR

G-Protein Coupled Receptor

7-transmembrane receptor
largest receptor family
metabotropic (not ion channel)

GENERAL GPCR STEPS (11-step cascade)

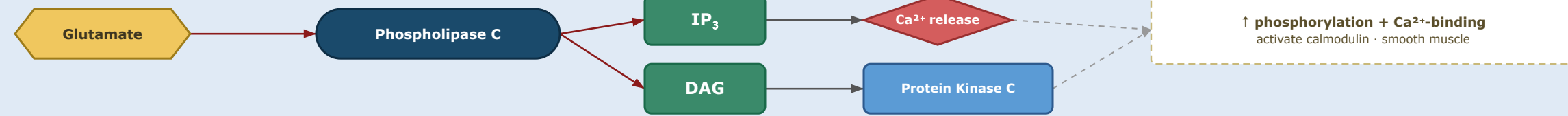


GPCR EFFECTOR PATHWAYS

G_s pathway



G_q pathway

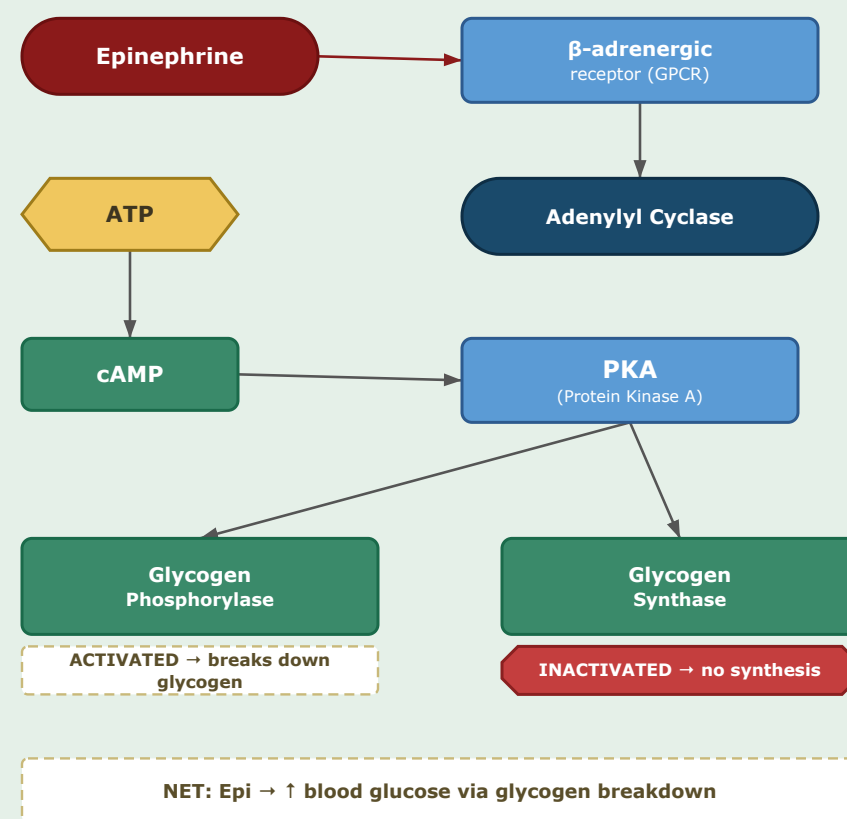


G_i pathway



KEY: G_s = stimulatory (↑cAMP) · G_q = PLC pathway (IP₃+DAG) · G_i = inhibitory (↓cAMP)
All use GTP hydrolysis as molecular switch · Cholera toxin locks G_s ON · Pertussis locks G_i OFF

EPINEPHRINE SIGNALING (example)



SUBSTRATES + GTP HYDROLYSIS

